Consumer Shopping Behavior Analysis Using Python, SQL, and Power BI

Abstract

Retail businesses generate large volumes of customer transaction data that can be leveraged to improve customer engagement, increase sales, and optimize business strategies. This project analyzes consumer shopping behavior data to identify purchasing patterns, customer preferences, and factors influencing buying decisions. Using Python for data cleaning and transformation, SQL for business analysis, and Power BI for visualization, the project provides actionable insights that support data-driven decision-making.

1. Introduction

1.1 Background

The retail industry is highly competitive, requiring businesses to understand customer behavior and preferences. Consumer shopping data contains valuable information about purchasing habits, demographics, product preferences, payment methods, and customer loyalty.

Analyzing this information helps organizations:

- Improve customer satisfaction
- Increase sales revenue
- Enhance marketing effectiveness
- Strengthen customer retention
- Optimize product offerings

1.2 Problem Statement

A retail company wants to better understand customer shopping behavior across product categories, demographics, payment methods, and promotional activities. The company seeks to identify factors that influence purchase decisions and customer loyalty in order to improve marketing strategies, customer engagement, and business performance.

1.3 Business Question

How can the company leverage consumer shopping data to identify purchasing trends, improve customer engagement, increase customer loyalty, and optimize marketing and product strategies?

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2. Objectives

The objectives of this project are:

1. Clean and prepare consumer shopping data.
2. Analyze customer purchasing behavior.
3. Identify high-performing product categories.
4. Evaluate the impact of discounts and subscriptions.
5. Understand customer demographics.
6. Build interactive dashboards for business stakeholders.
7. Provide business recommendations based on findings.

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3. Dataset Description

The dataset contains 3,900 customer records and 18 attributes.

Features

- Customer ID
- Age
- Gender
- Item Purchased
- Category
- Purchase Amount (USD)
- Location
- Size
- Color
- Season
- Review Rating
- Subscription Status

- Shipping Type
- Discount Applied
- Promo Code Used
- Previous Purchases
- Payment Method
- Frequency of Purchases

4. Data Cleaning and Preparation

4.1 Missing Values

The dataset contained missing values in the Review Rating column.

Action Taken

- Missing review ratings were replaced using the median rating value.

Python Example

```
df['Review Rating'].fillna(df['Review Rating'].median(), inplace=True)
```

4.2 Duplicate Records

Duplicate records were checked and removed to ensure data consistency.

```
df.drop_duplicates(inplace=True)
```

4.3 Data Type Validation

Data types were verified and corrected where necessary.

Examples:

- Age → Integer
- Purchase Amount → Numeric
- Review Rating → Float

4.4 Feature Engineering

Additional features were created for deeper analysis.

Age Groups

- Young
- Adult
- Middle-Age
- Senior

Customer Loyalty Indicator

Previous Purchases was used to estimate customer loyalty.

5. Exploratory Data Analysis

5.1 Customer Demographics

Gender Distribution

- Male customers represented the majority of purchases.
- Female customers contributed a smaller but significant share.

Age Analysis

Average customer age:

44 years

Major purchasing activity was concentrated among:

- Adults
- Middle-Age customers

5.2 Purchase Behavior

Average Purchase Amount:

$59.76

Purchase amounts ranged from:

- Minimum: $20
- Maximum: $100

This indicates relatively stable spending behavior.

5.3 Product Category Analysis

Four primary categories:

1. Clothing
2. Accessories
3. Footwear
4. Outerwear

Revenue by Category

Clothing generated the highest revenue.

Followed by:

- Accessories
- Footwear
- Outerwear

This indicates clothing products are the strongest revenue drivers.

5.4 Subscription Analysis

Subscription Status:

- No Subscription: 92.7%
- Subscription: 7.3%

Observation:

Most customers are non-subscribers, indicating significant opportunities to improve membership enrollment.

5.5 Review Rating Analysis

Average Review Rating:

3.76 / 5

This suggests moderate customer satisfaction.

Potential improvements:

- Product quality enhancement

- Customer support improvement
- Personalized recommendations

5.6 Payment Method Analysis

Popular payment methods included:

- PayPal
- Credit Card
- Debit Card
- Cash
- Venmo

Businesses should maintain support for multiple payment options to maximize convenience.

5.7 Shipping Analysis

Shipping methods included:

- Free Shipping
- Express
- Store Pickup
- Next Day Air
- 2-Day Shipping

Free Shipping was the most preferred option.

6. SQL Business Analysis

The cleaned dataset was imported into SQL for business intelligence analysis.

Query 1

Revenue by Category

Purpose:

Identify the highest revenue-generating product categories.

Finding

Clothing generated the highest revenue.

Query 2

Average Purchase Amount by Gender

Purpose:

Compare spending patterns between male and female customers.

Finding

Male customers contributed slightly higher overall revenue due to larger representation in the dataset.

Query 3

Discount Impact Analysis

Purpose:

Determine whether discounts influence spending behavior.

Finding

Discounts increased customer engagement and purchase frequency.

Query 4

Customer Loyalty Analysis

Purpose:

Identify repeat customers using Previous Purchases.

Finding

Customers with higher purchase histories generated significantly greater lifetime value.

Query 5

Payment Method Preferences

Purpose:

Analyze customer payment preferences.

Finding

Digital payment methods were among the most frequently used options.

7. Power BI Dashboard

An interactive dashboard was created to visualize consumer behavior.

Dashboard Components

KPI Cards

- Number of Customers
- Average Purchase Amount
- Average Review Rating

Revenue Analysis

- Revenue by Category
- Revenue by Age Group

Customer Analysis

- Gender Distribution
- Subscription Status
- Customer Segmentation

Sales Analysis

- Sales by Category

- Sales by Age Group

Filters

- Gender
- Category
- Subscription Status
- Shipping Type

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8. Key Insights

Insight 1

Clothing is the most profitable product category.

Insight 2

Average customer spending is approximately $60 per transaction.

Insight 3

Most customers are not subscribed to the membership program.

Insight 4

Middle-age and adult customers contribute the highest revenue.

Insight 5

Free Shipping remains the most preferred shipping option.

Insight 6

Customer ratings indicate moderate satisfaction levels with room for improvement.

9. Business Recommendations

Marketing

- Introduce personalized marketing campaigns.
- Promote subscription programs aggressively.
- Offer targeted promotions to repeat customers.

Product Strategy

- Expand clothing inventory.
- Improve low-rated products.
- Create product bundles.

Customer Retention

- Launch loyalty reward programs.
- Offer exclusive discounts to repeat customers.
- Implement customer retention campaigns.

Operational Improvements

- Maintain free shipping incentives.
- Improve delivery speed for premium customers.
- Enhance customer service support.

10. Conclusion

This project successfully analyzed consumer shopping behavior using Python, SQL, and Power BI. The findings reveal important trends related to customer demographics, purchasing behavior, product performance, and loyalty. Clothing emerged as the strongest revenue contributor, while subscription adoption remains relatively low. The dashboard and analytical insights provide management with valuable information to improve marketing effectiveness, increase customer retention, and drive business growth.

Future enhancements may include customer segmentation using machine learning, customer lifetime value prediction, recommendation systems, and churn prediction models.